

Sappa Jagadeesh

East Godavari, Andhra Pradesh

sappsjagadeesh@gmail.com | +91-9912493997 | [linkedin.com/in/jagadeesh-sappa](https://www.linkedin.com/in/jagadeesh-sappa) | github.com/jagadeesh08-s

Professional Summary

Electronics & Communication Engineering student with a unique blend of hardware expertise and AI-driven software development skills. Experienced in embedded systems, quantum circuit simulation, and utilizing Large Language Models (LLMs) to architect, generate, and debug complex applications rapidly. Recognized as a multi-hackathon winner (1st Prize Web Saga, 2nd Place AQVH) by leveraging strong system-level problem-solving and AI-assisted prototyping. Eager to contribute to forward-thinking teams building innovative technology solutions.

Education

BVC College of Engineering

B.Tech in Electronics & Communication Engineering

Rajahmundry, AP

2024 - 2027 (Expected)

Sir C.R. Reddy Polytechnic College

Diploma in Electronics & Communication Engineering | GPA: 88%

Eluru, AP

2021 - 2023

Technical Skills

AI & Software Workflow: AI-Assisted Development, Rapid Prototyping, Prompt Engineering, Git/GitHub, System Debugging

Programming (Foundational): Python, REST APIs, HTML/CSS, C

Electronics & Hardware: FPGA Architecture, VLSI Design, Digital Logic, Embedded Systems, Power Electronics, Signal Processing, Analog Circuits

Emerging Tech (Quantum): Qiskit, OpenQASM 2.0, IBM Quantum, Statevector Computation, Gate-Level Design

Projects

AI-Assisted Full-Stack Web Application | AI Prototyping, System Integration

Web Saga Hackathon

- Leveraged AI-assisted development tools to rapidly architect and generate a functional full-stack web application under strict hackathon time constraints.
- Acted as lead system integrator, reading, testing, and debugging AI-generated backend logic and frontend UI components to ensure seamless end-to-end functionality.
- Secured **1st Prize**, demonstrating strong system-level understanding, problem-solving, and the ability to execute complex technical projects using modern AI workflows.

E-RISHWA - Solar Powered Electric Auto | Power Electronics, Embedded Systems

Academic Project

- Designed a solar energy harvesting and integration system for an electric vehicle drivetrain.
- Optimized MPPT-based battery charging mechanism, successfully improving vehicle range by 30%.
- Applied practical power electronics concepts including DC-DC converter design and charge controller circuitry.

Bloch Verse - State Visualizer | Python, Qiskit, OpenQASM

AQVH 2025

- Built a real-time statevector extraction engine mapping simulated quantum states to interactive 3D visualizations.
- Integrated backend simulations with interactive dashboards to analyze multi-qubit states and error mitigation.
- Won **1st Place** at the national-level Amaravathi Quantum Valley Hackathon 2025 against 120+ teams.

Hardware Circuit Design Portfolio | Analog & Digital Electronics

Academic Project

- Engineered a regulated DC power supply incorporating op-amp based signal conditioning and filtering circuits.
- Constructed digital counter circuits using flip-flops and conducted frequency response analysis on RC networks.

Achievements & Publications

Hackathon Winner

2nd Place at National Level Hackathon among 120+ competing teams

Amaravathi Quantum Valley Hackathon (AQVH)

2025

Hackathon Winner

1st Prize in Web Saga Full-Stack Web Application Hackathon

GMR Institute of Technology

2024

Research Publication

Co-authored: "Design and Implementation of an Interactive Visualization Framework" (Peer-Reviewed)

APSHES Journal

2025